

Operational Readiness for LLM Hallucinations: A Counting-Process Capacity Model and Measured Evidence on Guardrail Cascades

Working draft. Revised from an earlier analytical-only draft; see Section 3.4 for corrections. Code and data: [hallucination_poisson/](#) in the accompanying repository.

Abstract

Organisations that deploy large language models (LLMs) must decide how much human triage capacity to hold in reserve and how many automated guardrail layers to place in front of the model. Practice relies on two rules of thumb: staff for the average incident load, and stack guardrails whose combined benefit is computed under independence. We model critical hallucinations as a (possibly over-dispersed, non-homogeneous) counting process, size capacity at a quantile of the daily count, and treat the multiplicative guardrail formula as the independence special case of Fréchet-type bounds. We show analytically that over-dispersion inflates required capacity by roughly the square root of the dispersion index, that positive association between layers makes the independence formula a lower bound on residual risk, and that, because capacity is integer-valued, total cost is not unimodal in the number of layers; an earlier claim that three layers are structurally optimal does not follow from the cost model. We then evaluate the integrated pipeline on measured outputs of three open LLMs answering 900 SQuAD 2.0 questions through three executed guardrail layers (rules, an extractive-QA classifier, an LLM judge) with measured latency and failures, with user traffic simulated as a non-homogeneous arrival process. We then test the Poisson structure on public production data: a request trace of an LLM service shows clustered arrivals at every scale above one second and a median day-to-day volume change of a factor of 1.8, and provider incident streams are strongly seasonal and remain over-dispersed after accounting for it. A single validated classifier cut required capacity by 80-95%; a routed judge added a modest, model-dependent gain at about triple the latency. The independence formula understated the share of hallucinations passing a two-layer cascade by a factor of 1.6-2.6 (2.7-4.6 with routing), a classifier with chance-level discrimination appeared beneficial only because it blocked almost everything, and with mild burstiness (daily-rate CV 0.3) a Poisson-sized capacity plan was exceeded on about 25-30% of days instead of 5%, while over-dispersion mattered little for low-rate provider-level incidents, as the thinning argument of Proposition 5 predicts. Hallucination arrival times are simulated and no public time-stamped hallucination log exists; models are small and the task is extractive QA, so the numbers do not transfer to production; the method and its diagnostics do.

Keywords: LLM operations, hallucination, guardrails, capacity planning, Poisson process, over-dispersion, design science

1. Introduction

Large language models (LLMs) are now embedded in customer-facing and decision-support information systems, where an ungrounded answer is a service incident rather than a quality blemish. IS research has examined adoption and value of generative AI, but the *operations* question that follows deployment has received little formal treatment: how much human triage and verification capacity must an organisation keep on standby, and how many automated guardrail layers should sit in front of the model, when hallucinations arrive at random?

Practitioners answer this with rules of thumb: staff for the average incident load, or stack as many guardrails as the latency budget tolerates. Both are costly mistakes. Staffing at the mean leaves the organisation under-provisioned on roughly one day in three to one in two, and the benefit of an additional guardrail is usually computed under an independence assumption that, as we show, is both unverified and optimistic.

We make three contributions, and we are explicit about which parts are new and which apply known results. **First**, we give an analytical capacity model for LLM incident handling in which critical hallucinations form a (possibly non-homogeneous, possibly over-dispersed) counting process, and the readiness requirement is the quantile of the daily count (Section 3). The quantile rule and the over-dispersion correction are classical (Whitt, 2007; Cameron and Trivedi, 1998); our contribution is to apply them to this arrival stream and to test whether the counting-process assumptions hold on measured LLM outputs. **Second**, we restate the multiplicative guardrail formula as the independence special case of Fréchet-type bounds on an arrival-rate (thinning) cascade (Proposition 3). Dependence between guardrail layers has been measured on real defence stacks (Alotaibi et al., 2026) and the independence product is known to over-credit redundancy (Bhardwaj et al., 2026); we add a measurement on hallucination-grounding layers and carry the correction through to the capacity decision. We then show that, once capacity is integer-valued, total cost is generally *not* unimodal in the number of layers and give comparative statics in the labour-to-compute cost ratio (Proposition 4); this follows from the model structure and is presented as a model property, not an empirical discovery. **Third**, we evaluate the integrated pipeline on real model outputs (three open models answering SQuAD 2.0 questions, three guardrail layers executed with measured latency and error behaviour) with traffic replayed through a non-homogeneous arrival process whose daily-rate variability is anchored in a public production trace of an LLM service (BurstGPT; Wang et al., 2025), and we test the Poisson assumptions directly on that trace and on the public incident history of three LLM providers (Chu et al., 2025). To our knowledge, this is the first study that tests counting-process assumptions on measured LLM hallucination outputs and connects the result to quantile-based triage capacity and guardrail-depth decisions in one framework; each ingredient has precedents (Section 2). Arrival times of the hallucination replay are simulated; outcomes, guardrail decisions and latencies are measured; the public traces concern request arrivals and service incidents, not hallucinations.

The paper follows the design-science tradition in IS (Hevner et al., 2004; Gregor and Hevner, 2013): we propose an artifact (a capacity-planning and guardrail-sizing method), ground it in a formal model, and evaluate it on real system outputs. We are explicit about what the evaluation does and does not establish (Section 6).

2. Related Work

Hallucination and its mitigation. Surveys document hallucination as a pervasive property of neural text generation (Ji et al., 2023). Mitigation work includes retrieval grounding (Lewis et al., 2020), sampling-based consistency checks (Manakul et al., 2023), safeguard classifiers (Inan et al., 2023) and LLM-as-judge verification (Zheng et al., 2023). This literature evaluates *detectors* by accuracy; it does not ask how detector performance translates into the human capacity an organisation must hold in reserve.

Capacity planning under random demand. The use of the Poisson quantile to size service capacity is classical in service operations (Whitt, 2007), and time-varying demand is handled by piecewise-stationary staffing (Green et al., 2007). Empirical studies of call centres report that arrivals are well described by a

Poisson process only conditional on a random, day-specific rate, i.e. they are over-dispersed (Brown et al., 2005); count-data econometrics offers the negative-binomial remedy (Cameron and Trivedi, 1998). We import these tools to a new arrival stream and test whether the same caveats apply.

Human review capacity for AI outputs. Queueing models of human review of AI output exist: Poisson escalation arrivals with Erlang-C staffing of human operators for agent escalations (Besanson, 2026), M/G/1 models of reviewing AI drafts (Bartolucci and Vivo, 2026), and fluid-limit routing between an LLM judge and human review (Lin and Zhang, 2026). These use mean-based stability or waiting-time criteria and assume, rather than test, Poisson arrivals. We size capacity at a quantile of the incident-count distribution and test the arrival assumptions.

Counting-process models of AI failures. Public-health-style monitoring of AI incident counts has found strong over-dispersion relative to Poisson in incident-database counts of vehicle and deepfake incidents (Abraham et al., 2026), and Markov reliability models of LLM agents reduce to the Goel-Okumoto NHPP software-reliability model in the rare-failure regime (Tran-Truong and Le, 2026). Neither concerns hallucination or guardrail-bypass events, nor applies time-rescaling tests or capacity sizing.

Layered defences. The multiplicative formula for the residual rate of independent safeguards is standard in reliability engineering, and it over-credits redundancy when failures are positively associated (Esary et al., 1967). For LLM defence stacks, Alotaibi et al. (2026) measure positive failure correlation between seven defence layers (ϕ between 0.30 and 0.75 on JailbreakBench) and call the product a lower bound that no real stack attains; Bhardwaj et al. (2026) give a Fréchet-Hoeffding sandwich and measure high co-failure among same-model LLM components. Our dependence results should be read as additional evidence, in a different setting (hallucination-grounding layers), rather than as a first demonstration.

Gap. Prior work addresses these pieces separately: arrival models for AI incidents, human-review queueing, measured layer dependence, and guardrail cost-latency trade-offs. We integrate them for LLM hallucination operations and validate the integrated pipeline on measured outputs. The literature on LLM guardrails moved quickly in 2026; our search was not exhaustive and similar work may exist.

3. Model

3.1 Arrivals of critical hallucinations

Users send queries at times governed by an intensity $\Lambda(t)$ with a daily cycle. Each query independently produces a *critical hallucination* (a wrong or ungrounded answer that passes all guardrails and reaches the user) with probability q . By the thinning theorem (Kingman, 1993), critical hallucinations form a non-homogeneous Poisson process (NHPP) with intensity $q\Lambda(t)$. Let X_d be the number of events on day d , with mean $\lambda = q \int_{\text{day}} \Lambda(t) dt$. Then $X_d \sim \text{Poisson}(\lambda)$ regardless of the within-day shape of Λ ; the within-day shape matters for *hourly* staffing, where the hour- h requirement is the quantile of $\text{Poisson}(\lambda_h)$.

Inter-arrival times are exponential only under a constant rate. Under the NHPP, the time-rescaled gaps $\Lambda(t_i) - \Lambda(t_{i-1})$ are i.i.d. $\text{Exp}(1)$ (the time-rescaling theorem), which gives a goodness-of-fit test that we use in Section 5. The memoryless corollary of the earlier draft of this work therefore applies to rescaled time: elapsed time since the last event carries no information, but time-of-day does.

If the daily rate is itself random, $\Lambda_d = \lambda M$ with $E[M] = 1$ and $\text{Var}(M) = c^2$, the daily count is a mixed Poisson variable with $\text{Var}(X_d) = \lambda + \lambda^2 c^2 = \lambda \phi$, where $\phi = 1 + \lambda c^2$ is the dispersion index. This is the Cox-process alternative to the pure Poisson null.

3.2 Readiness capacity

For reliability target α , the capacity requirement is the generalised inverse of the daily-count distribution:

$$k^*(\alpha, \lambda) = \min \{ k \in \mathbb{N}_0 : P(X_d \leq k) \geq \alpha \}$$

Proposition 1 (Poisson capacity). Let $X \sim \text{Poisson}(\lambda)$. (a) $k^*(\alpha, \lambda)$ is non-decreasing in λ . (b) $k^*(\alpha, \lambda) = 0$ if and only if $\lambda \leq -\ln \alpha$; for $\alpha = 0.95$ this threshold is 0.0513 events/day. (c) Provisioning at the mean, $k = \lfloor \lambda \rfloor$, leaves the system under-provisioned with probability $P(X > \lfloor \lambda \rfloor)$, which is at least 0.26 for every $\lambda \geq 1$, is about 0.35 at $\lambda = 3$ (not 0.42, which is $P(X \leq 2)$) and tends to 0.5 as λ grows. (d) $k^* = \lambda + z_{\alpha} \sqrt{\lambda} + O(1)$ as $\lambda \rightarrow \infty$.

Proof sketch. (a) Poisson(λ) is stochastically increasing in λ . (b) $P(X = 0) = e^{-\lambda} \geq \alpha$ iff $\lambda \leq -\ln \alpha$. (c) On $[n, n+1)$ the probability $P(X > n)$ is increasing in λ , so its minimum over $\lambda \geq 1$ is $P(X > 1) = 0.264$ at $\lambda = 1$, and it is 0.353 at $\lambda = 3$ ($P(X \leq 3) = 0.647$). (d) follows from the Cornish-Fisher expansion of the Poisson quantile. ■

Proposition 2 (over-dispersion inflates capacity). Let $X_d | M \sim \text{Poisson}(\lambda M)$ with $E[M] = 1$ and $\text{Var}(M) = c^2 > 0$. Then (a) $\text{Var}(X_d) = \lambda \phi$ with $\phi = 1 + \lambda c^2$; (b) X_d dominates Poisson(λ) in the convex order, so $E[(X_d - k)^+] \geq E[(Y - k)^+]$ for every k , where $Y \sim \text{Poisson}(\lambda)$; (c) $k^*_{\text{mixed}}(\alpha) = \lambda + z_{\alpha} \sqrt{\lambda \phi} + O(1)$, so the safety stock $z_{\alpha} \sqrt{\lambda}$ is inflated by the factor $\sqrt{\phi}$, and a plan sized with the Poisson quantile is exceeded on more than a $1 - \alpha$ share of days for large k .

Proof sketch. (a) is the law of total variance. (b) For Poisson, $\partial/\partial \lambda E[(Y - k)^+] = P(Y \geq k)$, which is non-decreasing in λ , hence $\lambda \mapsto E[(Y_{\lambda} - k)^+]$ is convex; Jensen's inequality applied to the mixing variable M gives $E[(X_d - k)^+] = E[g(\lambda M)] \geq g(\lambda)$. (c) follows by normal approximation with variance $\lambda \phi$. ■

Table 1 evaluates Proposition 2 exactly (negative-binomial mixing). At $\lambda = 10$ events/day and $c = 0.3$, the Poisson plan $k^* = 15$ is exceeded on 11.0% of days instead of 4.9%, and the correct requirement is 18; at $\lambda = 25$ and $c = 0.3$ the Poisson plan of 33 is exceeded on 16.7% of days and 41 is required.

Table 1. Capacity under over-dispersed arrivals (Gamma-mixed Poisson, $\alpha = 0.95$)

λ (events/day)	c (CV of daily rate)	ϕ	k^* Poisson	k^* mixed	P(exceed) Poisson plan
10	0	1.00	15	15	4.9%
10	0.1	1.10	15	16	5.7%
10	0.2	1.40	15	17	8.0%
10	0.3	1.90	15	18	11.0%
10	0.5	3.50	15	21	16.4%
25	0	1.00	33	33	5.0%
25	0.1	1.25	33	35	7.0%
25	0.2	2.00	33	37	11.9%
25	0.3	3.25	33	41	16.7%
25	0.5	7.25	33	50	23.0%

Exact negative-binomial quantiles; computed by `paper_tables.py`.

3.3 Thinning: guardrails make the incident stream closer to Poisson

Requests may be clustered even when incidents are rare. Because critical hallucinations are an independent thinning of requests, the dispersion of the incident stream follows from that of the request stream:

Proposition 5 (thinning reduces dispersion). Let N be the request count in a window with mean μ and Fano factor $F = \text{Var}(N)/\mu$, and let each request independently become a critical event with probability p . Then the event count N_p has mean $p\mu$ and Fano factor $1 + p(F - 1)$. In particular, a guardrail cascade that reduces the residual probability p pulls the incident stream towards Poisson (Fano factor 1), while the unprotected stream (p large) retains most of the request stream's burstiness. Equivalently, for a mixed-Poisson day with rate multiplier of CV c , $\phi = 1 + \lambda c^2$ in Proposition 2 with $\lambda = p\mu$.

Proof. $\text{Var}(N_p) = p^2\text{Var}(N) + p(1 - p)\mu$ by the law of total variance for the binomial thinning. Dividing by $p\mu$ gives $pF + (1 - p) = 1 + p(F - 1)$. ■ (We also verified this numerically by simulation.) The practical reading: over-dispersion matters most precisely where capacity is largest, i.e. without or with weak guardrails, and matters less after strong filtering, consistent with the shrinking gap between Poisson and negative-binomial capacity in Section 5.4.

3.4 Guardrail cascades and dependence

A cascade of N layers with stand-alone clearance rates c_i (the share of critical hallucinations layer i blocks) is usually assumed to give $\lambda_N = \lambda_0 \prod (1 - c_i)$. This presumes independent layer decisions. In practice the same ambiguous or adversarial outputs tend to fool all layers.

Proposition 3 (dependence bounds). For any joint behaviour of the layers, $\lambda_0(1 - \sum c_i)^+ \leq \lambda_N \leq \lambda_0(1 - \max_i c_i)$. If the layers' catch indicators, conditional on a hallucination, are positively associated (Esary et al., 1967), then additionally $\lambda_N \geq \lambda_0 \prod (1 - c_i)$. Consequently the multiplicative formula under-states residual risk and, through Proposition 1(a), under-states required capacity.

Proof sketch. The bounds are the Fréchet-Bonferroni bounds for the event that no layer fires. For associated indicators, $P(\cap \{I_i = 0\}) \geq \prod P(I_i = 0)$ (Esary et al., 1967). ■

Table 2 illustrates the width of the identification problem with $c_i = 0.63$ and $\lambda_0 = 10$: without knowing the dependence, three layers could imply anything from 0 to 7 units of capacity, which is why the dependence must be *measured* (Section 5.2).

Table 2. Residual rate and capacity bounds ($\lambda_0 = 10$, $c_i = 0.63$, $\alpha = 0.95$)

N	λ_N lower	λ_N independent	λ_N upper	k* lower	k* indep.	k* upper
1	3.70	3.70	3.70	7	7	7
2	0.00	1.37	3.70	0	4	7
3	0.00	0.51	3.70	0	2	7
4	0.00	0.19	3.70	0	1	7

3.5 Cost-minimising guardrail depth

Let C_h be the daily cost of one unit of triage capacity, g_i the daily compute cost of layer i , $k_{\min} \geq 0$ a minimum standing capacity (an on-call engineer is not removed because incidents are rare), and consider the cost

$$f(N) = C_h \cdot \max\{k_{\min}, k^*(\alpha, \lambda_N)\} + \sum_{i \leq N} g_i \quad (+ \text{ latency-feasibility: } \sum L_i \leq SLA).$$

Proposition 4 (optimal depth). (a) f is in general not unimodal in N : because k^* is integer-valued with plateaus, f can rise and then fall again. (b) With $k_{\min} \geq 1$, let N_0 be the first N with $k^*(\alpha, \lambda_N) \leq k_{\min}$; every $N > N_0$ is strictly dominated, so $N \leq N_0$. (c) Let $\theta = C_h / g$ with equal per-layer compute cost g . The minimiser $N(\theta)$ is non-decreasing in θ : when human triage is relatively expensive, deeper cascades are optimal (Topkis, 1978). (d) The latency constraint restricts N to a feasible prefix of the cascade.

Proof sketch. (a) is shown by the example in Table 3. (b) For $N > N_0$, labour cost is constant at $C_h k_{\min}$ while compute increases. (c) Write $f(N) = g[N + \theta K(N)]$ with K non-increasing; the difference $f(N+1) - f(N) = g[1 + \theta(K(N+1) - K(N))]$ is non-increasing in θ , so the largest minimiser is non-decreasing in θ by Topkis' monotone comparative statics. (d) is immediate. ■

Correction to the earlier draft. The earlier version of this paper reported Table 3 up to $N = 5$ and concluded that three layers are "optimal" with a cost rebound at $N = 5$. Extending the same table to $N = 6$ and 7 shows the cost falls again at $N = 6$ when k^* reaches zero ($\lambda_6 = 0.026 < -\ln 0.95$), so under $k_{\min} = 0$ the minimum is $N = 6$, and the table itself ranks $N = 4$ (\$250) below $N = 3$ (\$375). With a realistic floor of one on-call engineer ($k_{\min} = 1$) the minimum is $N = 4$ (\$250). The claim that three layers are structurally optimal is not supported by the model; the optimal depth is an outcome of the cost ratio, the clearance rates and the dependence structure.

Table 3. Daily cost by guardrail depth ($\lambda_0 = 10$, $c = 0.63$ independent, $C_h = \$150/\text{unit}$, $g = \$25/\text{layer}$)

N	λ_N	k^*	Labour	Compute	Total ($k_{\min} = 0$)	Total ($k_{\min} = 1$)
0	10.000	15	\$2,250	\$0	\$2,250	\$2,250
1	3.700	7	\$1,050	\$25	\$1,075	\$1,075
2	1.369	4	\$600	\$50	\$650	\$650
3	0.507	2	\$300	\$75	\$375	\$375
4	0.187	1	\$150	\$100	\$250	\$250
5	0.069	1	\$150	\$125	\$275	\$275
6	0.026	0	\$0	\$150	\$150	\$300
7	0.009	0	\$0	\$175	\$175	\$325

Minimum: $N = 6$ under $k_{\min} = 0$; $N = 4$ under $k_{\min} = 1$. Costs are the earlier draft's illustrative assumptions.

4. Empirical Design

4.1 Data, models and ground truth

We use the SQuAD 2.0 validation set (Rajpurkar et al., 2018), which combines answerable questions with questions that the supplied passage cannot answer. A random sample of 300 questions (seed 0) is drawn for each model. Each model receives the passage and the question and must reply with a short answer or the token UNANSWERABLE. A response is labelled a **critical hallucination** if it is not an abstention and

either (i) the question is answerable and the answer matches no gold answer (normalised containment or token-F1 ≥ 0.6) or (ii) the question is unanswerable. Abstentions are never counted as hallucinations and bypass the guardrails. The generators are Qwen2.5-0.5B-Instruct, SmolLM2-360M-Instruct and Qwen2.5-1.5B-Instruct (Qwen Team, 2024; Allal et al., 2025), run greedily on CPU. We use small open models because they can be run end-to-end with full control and measured latency; their hallucination rate is higher than that of frontier models, so absolute rates should not be transferred (Section 6).

4.2 Guardrail layers

The three layers follow the asymmetric architecture of the earlier draft, but each is implemented and executed on every answer:

- **L1, deterministic.** Regular expressions for e-mail, SSN, card and phone patterns; template-token leakage; degenerate repetition; and a grounding heuristic that blocks answers containing a number absent from the evidence passage.
- **L2, classifier.** A small NLI cross-encoder (DeBERTa-v3-small; He et al., 2021) scoring $P(\text{passage entails "the answer to Q is A"})$; the output is blocked if this probability is below 0.5. (The earlier draft named Llama Guard-type models for this layer; these target safety policy violations, not factual grounding, so an entailment classifier is the appropriate classifier for hallucinations.)
- **L3, semantic judge.** An LLM judge (Qwen2.5-1.5B-Instruct) given the passage, question and answer, replying SUPPORTED or UNSUPPORTED (Zheng et al., 2023). It is invoked only when routed: when L2 is present, only if the L2 entailment probability is below 0.9.

All layers use the same evidence passage as retrieval-grounded verification. Every layer is run on every example so that any subset or cascade can be evaluated offline from observed flags, without any independence assumption. We record flag, score, wall-clock latency and exceptions per call. Layers fail open; a call that errors or exceeds a 2 s timeout is counted as a system failure, from which we estimate $\lambda_{\text{sys_fail}}$ rather than assuming it.

4.3 Traffic replay and estimators

Production logs are unavailable, so *arrivals* are simulated while *outcomes* are measured. Queries arrive according to an NHPP with a 24-slot hour-of-day intensity and a mean of 200 queries/day for 90 days. The default profile is business-hours-shaped; a second profile is the hour-of-day profile measured in the BurstGPT production trace (Section 4.5). A Cox variant multiplies each day's intensity by a Gamma factor with mean 1 and coefficient of variation c to represent bursty demand; we use $c = 0.3$ (an arbitrary mild scenario) and $c = 0.44$ (the coefficient of variation of the calmest 14-day window of BurstGPT, Section 5.6, which makes the scenario data-anchored and still conservative). Each arriving query replays a uniformly drawn example from the measured pool; it becomes a critical event if it is a hallucination that survives the cascade. For each resulting event stream we estimate $\hat{\lambda} = \text{events/days}$ and test: (i) the index-of-dispersion test on daily counts; (ii) a chi-square test of constant hourly intensity; (iii) a Kolmogorov-Smirnov test of raw inter-arrival times against an exponential, and of time-rescaled gaps under the fitted hourly intensity (the Lilliefors-type p-value is optimistic); (iv) a negative-binomial fit of daily counts. Uncertainty from the finite evaluation pool is quantified by a 300-draw bootstrap over examples (95% percentile intervals).

Because the same pool generates all arrivals, the replay tests whether the *counting process* has the claimed structure given measured per-query outcomes; it does not test whether real users' queries are independent. That is the main external-validity limit (Section 6).

4.4 Cost model

Daily cost is $C_h \cdot \max(k_{\min}, k^*) + \text{compute} + \text{false-block cost} + \text{residual-harm cost}$. We set $C_h = \$150$ per capacity unit (as in the earlier draft), compute \$0.0001 per classifier call and \$0.002 per judge call, and treat the false-block cost c_{fp} and residual-harm cost c_{res} as scenario parameters in a sensitivity grid. These are assumptions, not estimates, and are labelled as such wherever used.

4.5 Public production data

Two public datasets test the counting-process assumptions on real systems; neither concerns hallucinations. **BurstGPT** (Wang et al., 2025; CC BY 4.0) contains 10.6 million request records of regional Azure OpenAI ChatGPT and GPT-4 services at one-second resolution over 231 days (two contiguous segments of 121 and 110 days separated by an unrecorded gap of 104 days; timestamps are seconds from an undisclosed day 0, so weekday labels are unknown); failed requests are the rows with zero response tokens. We estimate the hour-of-day profile, the day-to-day variability of request volume (robustly, because volume grows by two orders of magnitude and contains regime shifts), and the within-hour Fano factor of counts at 1 s, 10 s, 1 min and 10 min. **LLM Service Outages and Incident Reports** (Chu et al., 2025; CC BY 4.0) contains provider-declared incidents scraped from the OpenAI, Anthropic and Character.AI status pages; we use the 457 incidents starting from March 2023, when all three pages are populated, and test Poisson counts per day and week (index of dispersion, also stratified by day of week), day-of-week and hour-of-day homogeneity, exponential inter-arrival times after time-rescaling with a fitted hour-of-week intensity, the distribution of incident durations as a proxy for human response time, and the number of concurrently open incidents against the $M/G/\infty$ prediction.

5. Results

5.1 Hallucination rates and guardrail layer performance

On 300 SQuAD 2.0 questions per model, the critical-hallucination rate is 60.3% (Qwen2.5-0.5B), 65.0% (SmolLM2-360M) and 47.0% (Qwen2.5-1.5B). Most hallucinations are answers to unanswerable questions. Table 4 reports the layers. **Layer L1** (rules) catches 0.7-5.1% of hallucinations: this task contains no PII or template leaks and few fabricated numbers, so the deterministic layer is nearly inert here, though free of latency and false blocks. **Layer L3** (the 1.5B judge) blocks 63-84% of hallucinations but also 25-39% of correct answers, and each call takes about 1.9 s on our CPU, with 25-30% of calls exceeding the 2 s timeout.

Negative result for the NLI classifier. Our first implementation of L2 was the NLI cross-encoder of the earlier draft. It was essentially uninformative: its entailment score separates correct from hallucinated answers with AUC 0.52-0.58 (0.5 is chance), and at the default threshold it blocked 87-88% of hallucinations and 85-90% of *correct* answers. Its apparent capacity savings were therefore an artefact of blocking nearly everything. We replaced it by an extractive-QA verifier (a SQuAD 2.0-trained RoBERTa model whose own answer span is compared with the generator's answer by token-F1; AUC 0.84-0.90) and re-ran all layers on the stored answers. This verifier is specialised to the task family (it was trained on

SQuAD), which flatters it relative to a generic classifier; all numbers below use it, and we keep the NLI results to show that a "classifier layer" is not automatically a useful one.

Table 4. Guardrail layers on 300 SQuAD 2.0 questions per generator (CPU inference, latency re-measured without concurrent jobs)

Generator	p_0 (hallucination rate)	c_1 (L1)	c_2 (L2 QA)	c_3 (L3 judge)	L2 false-block (good answers)	L3 false-block	L2 mean ms	L3 mean ms
Qwen2.5-0.5B	60.3%	2.8%	89.5%	84.0%	15.1%	39.5%	98	1941
SmolLM2-360M	65.0%	5.1%	96.9%	63.1%	69.5%	24.8%	97	1946
Qwen2.5-1.5B	47.0%	0.7%	81.6%	78.7%	13.2%	33.3%	95	1898

c_i = stand-alone share of hallucinated answers the layer blocks. False-block = share of correct (non-hallucinated, non-abstained) answers blocked. The L2 threshold ($F1 < 0.5$) was not tuned; SmolLM2 gives longer answers, which lowers its token-F1 agreement and inflates its L2 false-block rate.

5.2 Dependence between layers: the independence formula under-states residual risk

Among hallucinated answers, the catch decisions of L2 and L3 are positively correlated ($\phi = 0.15$ - 0.24 ; L1 pairs: $\phi \leq 0.09$), and the observed joint catch rate exceeds the independence prediction in every pair for every model, in the direction of Proposition 3. The consequence for the residual rate is large (Figure 2, left). For the L2+L3 cascade, the share of hallucinations that pass all layers is 1.1-3.9% under the independence formula, but 2.6-6.4% when both layers are run on every answer (1.6-2.6 times higher), and 3.1-15.6% with dynamic routing of only low-agreement answers to the judge (2.7-4.6 times higher). Routing therefore has a first-order cost that independence-based sizing hides: it reduces judge calls by about 89%, but the judge no longer sees answers that the verifier considers consistent, and those contain a substantial share of the hallucinations.

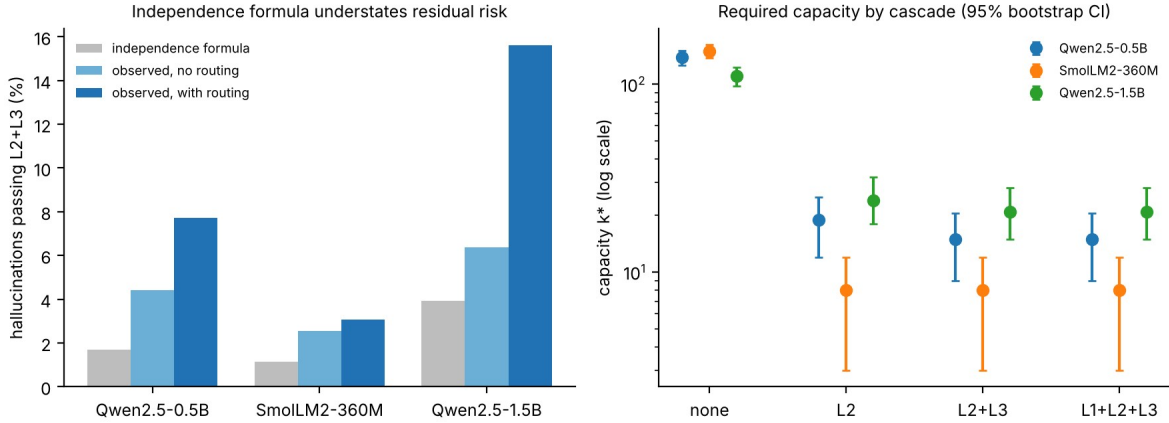


Figure 2. Left: share of hallucinations passing the L2+L3 cascade under independence, observed without routing, and observed with routing. Right: capacity k^* by cascade (95% bootstrap CI over the evaluated pool); log scale.

5.3 Counting-process checks under replayed traffic

With non-homogeneous Poisson arrivals and no burstiness, the replay reproduces the Poisson structure, as it must by construction: for the unprotected stream, the daily-count dispersion test does not reject ($p = 0.53$ - 0.91), the raw inter-arrival times are strongly non-exponential ($p < 10^{-100}$) because the intensity varies with the hour (hour-of-day homogeneity rejected), and the *time-rescaled* gaps are consistent with $\text{Exp}(1)$ ($p =$

0.50-0.85). We stress that this is a check that the estimators and the replay are correct and that the NHPP/rescaling machinery behaves as theory predicts; it is not evidence that real user traffic is a Poisson process, because the arrivals are simulated (Section 4.3). Within-day non-homogeneity matters for staffing by hour but not for the daily count.

5.4 Burstiness: Poisson capacity plans fail

When the daily intensity is multiplied by a Gamma factor with a coefficient of variation of only 0.3, the daily-count dispersion test rejects overwhelmingly ($p < 10^{-90}$ for the unprotected stream) and the negative-binomial fit gives r between 12.7 and 13.4. The Poisson capacity $k^* = 136, 146, 108$ (for the three generators, no guardrails) is then exceeded on 23-28% of days instead of the targeted 5%; the correct negative-binomial requirement is 178, 194, 141, i.e. 31-33% more (Figure 1; Proposition 2). Under the best cascade (L1+L2, $\lambda \approx 4$ -17/day) the effect is smaller in absolute capacity but still raises the exceedance probability of the Poisson plan to 7-10%. The value $c = 0.3$ is a scenario, not an estimate: the point is that a mild day-to-day variation in demand, which any real service has, is enough to invalidate the Poisson plan, so the dispersion must be estimated from production incident logs before capacity is fixed.

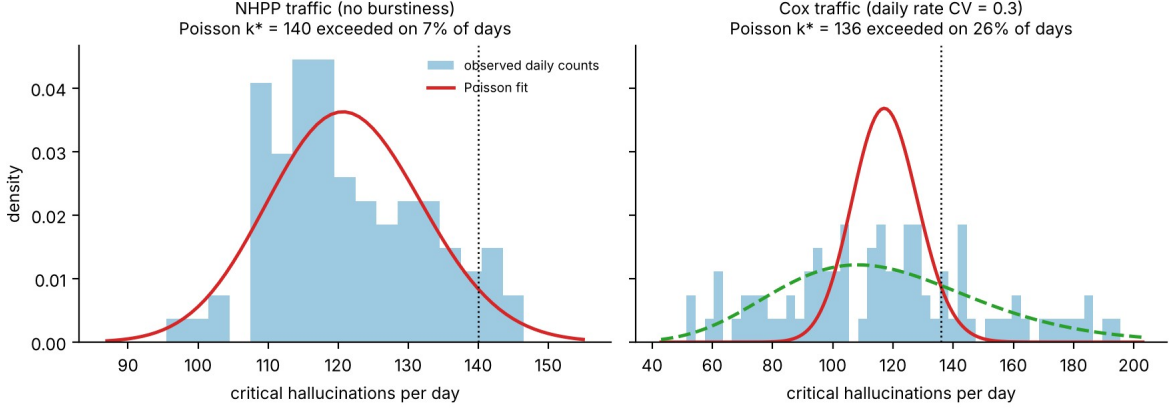


Figure 1. Daily critical hallucinations without guardrails (Qwen2.5-0.5B, 90 days, 200 queries/day). Left: steady NHPP traffic; right: bursty traffic. Dotted line: Poisson k^* ($\alpha = 0.95$).

Data-anchored scenario. Section 5.6 shows that real LLM request volume varies much more from day to day than $c = 0.3$; the calmest 14-day window of the BurstGPT trace still has $c = 0.44$. Replaying our measured outcomes with $c = 0.44$ and the BurstGPT hour-of-day profile gives the exceedance rates in Table 6.

Table 6. Share of days on which the Poisson-sized capacity k^* is exceeded (target 5%; 90 simulated days)

Generator	Cascade	Steady NHPP ($c = 0$)	$c = 0.3$	$c = 0.44$ + BurstGPT profile	k^* Poisson $\rightarrow$ k^* negative-binomial ($c = 0.44$)
Qwen2.5-0.5B	none	6.7%	25.6%	32.2%	135 $\rightarrow$ 209
Qwen2.5-0.5B	L1+L2	3.3%	10.0%	14.4%	18 $\rightarrow$ 23
SmolLM2-360M	none	5.6%	27.8%	32.2%	144 $\rightarrow$ 225
SmolLM2-360M	L1+L2	5.6%	11.1%	6.7%	8 $\rightarrow$ 9
Qwen2.5-1.5B	none	8.9%	23.3%	30.0%	106 $\rightarrow$ 162
Qwen2.5-1.5B	L1+L2	3.3%	10.0%	15.6%	24 $\rightarrow$ 32

Observed over 90 simulated days, so each share has a standard error of about 2-5 percentage points. L1+L2 is the cheapest cascade within the 500 ms budget (Section 5.5).

The gap shrinks with guardrails but does not vanish: for the best cascade, the Poisson plan is exceeded on 7-16% of days at $c = 0.44$, against 30-32% without guardrails. This is the thinning effect of Proposition 5: at small λ the product λc^2 is small, so over-dispersion inflates the safety stock less in absolute terms (about 5, 1, 8 extra units) although the exceedance probability is still several times the target.

5.5 Capacity, latency and cost by cascade

Table 5 gives, for each generator, the residual rate λ , the capacity k^* , the added latency and the false-block volume of the main cascades at 200 queries/day, with bootstrap intervals over the evaluated pool. Intervals are wide at this pool size (300 questions), so differences of a few units of k^* between cascades are not significant; the large differences are.

Table 5. Cascade performance at 200 queries/day, $\alpha = 0.95$, SLA 500 ms mean added latency (95% bootstrap CI in brackets)

Generator	Cascade	λ per day	k^*	Mean added latency (ms)	False blocks / day	System failures / day
Qwen2.5-0.5B	none	120.7 [109, 132]	139 [126, 151]	0	0.0	0.0
Qwen2.5-0.5B	L2	12.7 [7, 18]	19 [12, 25]	98	12.0	0.0
Qwen2.5-0.5B	L3	19.3 [12, 25]	27 [18, 34]	1941 $\times$	31.3	60.0
Qwen2.5-0.5B	L2+L3	9.3 [5, 14]	15 [9, 21]	311	17.3	6.0
SmolLM2-360M	none	130.0 [120, 142]	149 [138, 162]	0	0.0	0.0
SmolLM2-360M	L2	4.0 [1, 7]	8 [3, 12]	97	48.7	0.0
SmolLM2-360M	L3	48.0 [38, 57]	60 [48, 70]	1933 $\times$	17.3	50.0
SmolLM2-360M	L2+L3	4.0 [1, 7]	8 [3, 12]	271	50.7	6.7
Qwen2.5-1.5B	none	94.0 [82, 106]	110 [97, 123]	0	0.0	0.0
Qwen2.5-1.5B	L2	17.3 [12, 24]	24 [18, 32]	88	14.0	0.0
Qwen2.5-1.5B	L3	20.0 [13, 26]	28 [19, 35]	1752 $\times$	35.3	49.3
Qwen2.5-1.5B	L2+L3	14.7 [9, 21]	21 [15, 28]	296	22.0	6.7

$\times$ = violates the 500 ms budget. System failure = a layer call that errors or exceeds 2 s. $L1+L2$ and $L1+L2+L3$ equal $L2$ and $L2+L3$ up to small differences because $L1$ is nearly inert on this task.

Capacity. A single well-chosen classifier reduces required capacity from 139, 149, 110 to 19, 8, 24 units/day (a 78-95% reduction). Adding the routed judge reduces it further only modestly (15, 8, 21) and for SmolLM2 not at all, while multiplying added latency by about three and adding roughly 6 system failures per day through judge timeouts. Running the judge on every answer without routing is infeasible under any latency budget we consider (about 1.9 s) and produces about 50-60 system failures per day on our CPU setup.

Latency budget. Under the 100 ms budget of the earlier draft, only $L1$ and $L2$ ($L2$ takes about 90-100 ms here) are feasible and no judge layer fits; under 500 ms the routed judge fits on average but not at the 95th percentile (about 1.5 s). Latencies are CPU-bound and would be much lower on GPUs or hosted inference.

Cost. With $C_h = \$150$ per capacity unit, \$1 per falsely blocked answer and \$100 per hallucination reaching a user (illustrative scenario values), the cheapest cascade is $L1+L2+L3$ for Qwen2.5-0.5B, $L1+L2$ for SmolLM2-360M, $L1+L2+L3$ for Qwen2.5-1.5B; daily cost falls from \$32,917, \$35,350, \$25,900 without guardrails to \$3,201, \$1,649, \$4,639. The ranking is the same across the traffic $\times$ false-block-cost $\times$ harm-cost grid of Section 4.4 (27 scenarios per generator), but the dollar levels are assumptions. Whether the third

layer pays for itself depends on the generator: it does for Qwen2.5-0.5B and Qwen2.5-1.5B and does not for SmolLM2-360M, so a universal "three layers" prescription is not supported by these data. The classifier is the dominant layer, the deterministic layer is nearly free but also nearly useless on this task, and the judge's marginal value is positive but small and uncertain.

5.6 Evidence from public production traces

Request arrivals (BurstGPT). The Azure OpenAI trace has 10.6 million requests over 231 recorded days; 4.6% of requests failed (zero response tokens). The hour-of-day profile has a peak-to-trough ratio of 2.2 (peak in the late afternoon and evening hours, trough in the morning hours of the local clock), so the arrival process is non-homogeneous, as the NHPP formulation assumes. Within an hour, however, counts are far more variable than Poisson: the variance-to-mean ratio of counts per bin (Poisson = 1) has a median of 1.4 at 1 s, 5.3 at 10 s, 3.9 at 1 min and 5.1 at 10 min, i.e. requests arrive in clusters at every scale above one second. By Proposition 5 an unprotected hallucination stream with $p = 0.6$ would inherit about $1 + 0.6(F - 1) \approx 3.5$ of that dispersion at the 10-minute scale, whereas a cascade with residual probability $p = 0.05$ would retain about 1.2.

Day-to-day volume is extremely variable: the median day has 19,024 requests, with a range from 52 to 425,604. The median absolute day-to-day log change is 0.61 (a factor of 1.8), 73% of days differ from the previous day by more than 30%, the robust log-standard deviation around a 15-day rolling median is 0.80, and even the calmest 14-day window has a coefficient of variation of 0.44 (mean 140k requests/day). This variability reflects traffic growth, campaigns, regime shifts and possibly logging gaps and not only random fluctuation (the series also contains 7 near-duplicate day pairs, which we treat as a data artefact), so we use it as a **stress test and as an anchor showing that $c = 0.3$ is a conservative scenario**, not as an estimate of the dispersion of hallucination counts. As a stress test, replaying the measured daily multipliers on an incident stream of $\lambda = 10$ and 120 events/day makes the Poisson capacity plan fail on 21-30% of days (target 5%), with negative-binomial requirements of 33, 384, 75, 892 units against Poisson plans of 15, 138, 15, 138.

Service incidents (LLM providers). Between March 2023 and August 2024 the three providers declared 457 incidents (0.83 per day; OpenAI 0.51, Anthropic 0.26). Incident starts are strongly non-homogeneous: weekday counts are 2.5-6.4 times weekend counts and hour-of-day homogeneity is rejected ($p < 10^{-4}$ for every provider and subset). A homogeneous Poisson model is rejected on every inter-arrival and dispersion test. Allowing for weekly and daily seasonality removes part of the lack of fit but not all of it: the daily index of dispersion is 1.53 ($p = 3 \times 10^{-14}$) and still 1.30 after stratifying by day of week ($p = 6 \times 10^{-6}$); for OpenAI alone the time-rescaled inter-arrival test is consistent with an NHPP ($p = 0.23$) but its stratified index is still 1.19 ($p = 0.003$), while for Anthropic and for the major and critical subset both are rejected. A negative-binomial fit to daily counts gives $r \approx 1.6$. The practical consequence at this low rate is small: the Poisson and negative-binomial 95% capacities coincide (3 incidents per day) and the Poisson capacity is exceeded on 3.5% of days, below the 5% target. This agrees with Proposition 2: over-dispersion is statistically clear but $\phi = 1 + \lambda c^2$ stays near 1 when λ is below one per day, and it becomes consequential only at higher rates.

Human response proxy. Incident duration (declared start to resolution) has a median of 1.0 h, a 90th percentile of 5.1 h and a 99th percentile of 34 h; the mean is 3.1 h and the coefficient of variation 3.5, i.e. response times are heavy-tailed, not exponential (12.3% are zero-duration notices). Treating incidents as arrivals to an infinite-server system, the time-averaged number of concurrently open incidents is 0.11, matching the $M/G/\infty$ mean (Little's law), and the 95th percentile of the open count is 1, equal to the Poisson

95% quantile of the $M/G/\infty$ occupancy (1); two or more incidents are open simultaneously 0.7% of the time. At provider level a single standing responder therefore covers 95% of the time, whereas the heavy tail of durations explains the rare long overruns. These are provider-declared outage incidents (elevated errors, latency, degraded service), not hallucinations, so they inform the arrival and service-time structure of an LLM incident stream but not its rate.

6. Discussion

6.1 Implications for IS practice and research

- **Size capacity from a quantile, and fit the dispersion first.** Provisioning at the mean load fails on roughly 40-50% of days in every configuration we measured. The Poisson quantile fixes this when daily counts are Poisson, but Proposition 2 and the burst experiments show that a modest day-to-day variation in demand (CV 0.3-0.44) multiplies the miss rate several-fold; real LLM request streams vary far more than that. Dispersion should be estimated from incident logs before the plan is fixed, and over-dispersion matters most for large incident streams (weak guardrails) and least for small ones (Proposition 5): for provider-level outages of about 0.8 per day, over-dispersion was statistically clear but changed the 95% capacity by nothing.
- **Guardrail value is an empirical, per-deployment quantity.** The layers' standalone clearance, their dependence, their false-block cost and their latency vary strongly by layer and model. In our data one well-chosen classifier delivers most of the benefit, the deterministic layer is nearly inert, and the judge adds a modest, model-dependent gain at triple the latency. A fixed "three layers" prescription is not supported; the method here selects the stack from measurements. A classifier that is not validated can be actively harmful: our NLI layer had chance-level discrimination and appeared to cut capacity only because it blocked almost everything.
- **Routing is a first-order design choice.** Sending only low-agreement answers to the expensive judge cuts judge calls by roughly an order of magnitude, but it also forgoes catches among answers the classifier rated as consistent; routing raised the residual hallucination share by a factor of 1.2-2.4 relative to running both layers on everything (Section 5.2).
- **Governance.** Because k^* converts a detector's precision/recall into headcount, the framework gives IT governance a common unit (units of triage capacity or dollars per day) in which to compare detectors, set the SLA and justify standing on-call capacity.

6.2 Limitations

- **Simulated arrivals and no hallucination log.** Outcomes are measured on real model outputs, but arrival times are simulated and each arrival replays a bootstrapped example. The public traces we use (BurstGPT, the provider incident history) are real production data, but they record request arrivals and service incidents, not hallucinations; no public time-stamped production log of hallucination incidents exists to our knowledge. The burstiness of the replay is therefore a scenario anchored in, not estimated from, the dispersion of real request streams. Validation on proprietary incident logs is the main follow-up.
- **Public-trace artefacts.** BurstGPT has no calendar dates, a 104-day gap, near-duplicate day pairs and order-of-magnitude regime changes in daily volume, so its day-level variability reflects growth and logging as well as randomness; we use it as a stress test and a lower bound for c , not as an

estimate. The provider incident data are provider-declared, with minute resolution and detection and reporting bias; incident duration is the declared start-to-resolution time, not triage effort.

- **Small open models and one task.** We study 0.4-1.5B-parameter models on extractive QA with unanswerable questions. Absolute hallucination rates are much higher than for frontier models, and the layers are CPU-bound; latencies on GPU or hosted APIs differ. The *method* transfers, the *numbers* do not.
- **Label noise.** Ground truth uses string and token-F1 matching against SQuAD gold answers, which can mark valid paraphrases as hallucinations; this affects all configurations alike but inflates the base rate.
- **Cost parameters are assumptions.** Only labour-to-compute ratios and the sensitivity grid are informative; dollar levels are illustrative.
- **Single judge and classifier, and a post-hoc layer change.** Layer dependence and clearance were measured for one extractive-QA verifier and one judge model; other choices may be more or less correlated. The QA verifier was introduced after the NLI classifier proved uninformative, so the L2 choice was made after seeing results (on the same data); it is trained on SQuAD, which flatters it relative to a domain-agnostic classifier, and its threshold (token-F1 < 0.5) was not tuned. The analyses should be repeated on a held-out sample with a pre-registered layer configuration.
- **Consistency checks are partly by construction.** Because outcomes are bootstrapped from a fixed pool and arrivals are simulated, the Poisson and time-rescaling checks of Section 5.3 verify the machinery, not real traffic; only the dependence, clearance, latency and failure measurements are evidence about real systems.
- **Independent days.** The model treats days as exchangeable; weekly seasonality and trends are not modelled.

7. Conclusion

Treating critical hallucinations as a counting process turns guardrail performance into a staffing and cost decision. The analysis yields three results that differ from the common practice the earlier draft followed: capacity must be sized from an over-dispersion-aware quantile; the multiplicative guardrail formula is an optimistic bound whose gap must be measured; and the cost-minimising number of layers is not fixed at three but depends on integer capacity effects, the floor on standing staff, latency, the labour-to-compute ratio and, empirically, on the generator. On measured outputs of three open models, a single task-validated classifier reduces the required capacity by roughly 80-95% relative to no guardrails; a routed judge adds a further modest, model-dependent reduction at about triple the latency, and every cascade incurs false blocks and system failures that the earlier framework ignored. The independence formula understated the share of hallucinations passing a two-layer cascade by a factor of 1.6-2.6 (2.7-4.6 with routing), and a mild burstiness (daily-rate CV of 0.3) raised the exceedance rate of a Poisson-sized plan from 5% to about 25%, to about 30% with the CV of 0.44 anchored in a production trace. Public production data corroborate the structure: an LLM request trace shows clustered arrivals at every scale above one second and a median day-to-day volume change of a factor of 1.8, and provider incident streams are strongly seasonal and remain over-dispersed after accounting for seasonality, with heavy-tailed durations. Validating the counting-process assumptions on production incident logs is the key next step.

References

- Abraham, S., et al. (2026). AI incident monitoring through a public health lens. arXiv:2604.19914.
- Allal, L. B., et al. (2025). SmolLM2: When Smol goes big, data-centric training of a small language model. arXiv:2502.02737.
- Alotaibi, A., Jabbar, M. S., Al-Azani, S., & Ahmed, M. (2026). Layered LLM defenses as an ensemble: Access tiers, inference cost, and the measured failure correlation between defense layers. arXiv:2608.28327.
- Bartolucci, & Vivo, (2026). Queue & AI: When faster tasks slow down the workflow. arXiv:2605.27202.
- Besanson, (2026). SARC: A governance-by-architecture framework for agentic AI systems. arXiv:2605.07728.
- Bhardwaj, V. P., Singh, & Bhardwaj, (2026). Agent behavioral contracts II: Certifying compositional reliability without assuming independence. arXiv:2608.12895.
- Brown, L., Gans, N., Mandelbaum, A., Sakov, A., Shen, H., Zeltyn, S., & Zhao, L. (2005). Statistical analysis of a telephone call center: A queueing-science perspective. *Journal of the American Statistical Association*, 100(469), 36-50.
- Cameron, A. C., & Trivedi, P. K. (1998). *Regression Analysis of Count Data*. Cambridge University Press.
- Chu, X., Talluri, S., Lu, Q., & Iosup, A. (2025). An empirical characterization of outages and incidents in public services for large language models. *Proceedings of ICPE 2025* (arXiv:2501.12469).
- Cox, D. R. (1955). Some statistical methods connected with series of events. *Journal of the Royal Statistical Society, Series B*, 17(2), 129-164.
- Esary, J. D., Proschan, F., & Walkup, D. W. (1967). Association of random variables, with applications. *Annals of Mathematical Statistics*, 38(5), 1466-1474.
- Green, L. V., Kolesar, P. J., & Whitt, W. (2007). Coping with time-varying demand when setting staffing requirements for a service system. *Production and Operations Management*, 16(1), 13-39.
- Gregor, S., & Hevner, A. R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, 37(2), 337-355.
- He, P., Gao, J., & Chen, W. (2021). DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. arXiv:2111.09543.
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75-105.
- Inan, H., et al. (2023). Llama Guard: LLM-based input-output safeguard for human-AI conversations. arXiv:2312.06674.
- Ji, Z., et al. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), Article 248.
- Kingman, J. F. C. (1993). *Poisson Processes*. Oxford University Press.
- Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33.
- Lin, R., & Zhang, J. (2026). When to screen, when to bypass: LLM-judges in resource-scarce AI-human workflow. arXiv:2603.13870.
- Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. *Proceedings of EMNLP 2023*.
- Qwen Team. (2024). Qwen2.5 technical report. arXiv:2412.15115.
- Rajpurkar, P., Jia, R., & Liang, P. (2018). Know what you don't know: Unanswerable questions for SQuAD. *Proceedings of ACL 2018*, 784-789.
- Topkis, D. M. (1978). Minimizing a submodular function on a lattice. *Operations Research*, 26(2), 305-321.
- Tran-Truong, & Le, (2026). Measuring the unmeasurable: Markov chain reliability for LLM agents. arXiv:2604.24579.
- Wang, Y., et al. (2025). BurstGPT: A real-world workload dataset to optimize LLM serving systems. *Proceedings of KDD 2025* (arXiv:2401.17644).

- Whitt, W. (2007). What you should know about queueing models to set staffing requirements in service systems. *Naval Research Logistics*, 54(5), 476-484.
- Zheng, L., et al. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems*, 36.